PHP memory_limit remote vulnerability

PHP memory_limit remote vulnerability - Stefan Esser, e-matters GmbH.

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Content

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'Advisory 11/2004: PHP memory_limit remote vulnerability' - MARC

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List: [bugtraq](https://marc.info/?l=bugtraq&r=1&w=2)
Subject: Advisory 11/2004: PHP memory_limit remote vulnerability
From: [Stefan Esser <s.esser () e-matters ! de>](https://marc.info/?a=103331475800004&r=1&w=2)
Date: [2004-07-13 22:53:29](https://marc.info/?l=bugtraq&r=1&w=2&b=200407)
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-----BEGIN PGP SIGNED MESSAGE-----
Hash: SHA1

                          e-matters GmbH
                         www.e-matters.de

                      -= Security Advisory =-

     Advisory: PHP memory_limit remote vulnerability
 Release Date: 2004/07/14
Last Modified: 2004/07/14
       Author: Stefan Esser [s.esser@e-matters.de]

  Application: PHP <= 4.3.7
               PHP5 <= 5.0.0RC3
     Severity: A vulnerability within PHP allows remote code
               execution on PHP servers with activated memory_limit
         Risk: Critical
Vendor Status: Vendor has released a bugfixed version.
    Reference: http://security.e-matters.de/advisories/112004.html

Overview:

   PHP is a widely-used general-purpose scripting language that is
   especially suited for Web development and can be embedded into HTML.

   According to Security Space PHP is the most popular Apache module
   and is installed on about 50% of all Apaches worldwide. This figure
   includes of course only those servers that are not configured with

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   expose_php=Off.

   During a reaudit of the memory_limit problematic it was discovered
   that it is possible for a remote attacker to trigger the memory_limit
   request termination in places where an interruption is unsafe. This
   can be abused to execute arbitrary code on remote PHP servers.

Details:

   On the 28th June 2004 Gregori Guninski released his advisory about
   a possible remote DOS vulnerability within Apache 2 (CAN-2004-0493).
   This vulnerability allows tricking Apache 2 into acception arbitrary
   sized HTTP headers. Guninski and many others rated this bug as "Low
   Risk" for 32bit systems, but they did not take into account that
   such a bug could have a huge impact on 3rd party modules.

   After his advisory was released I reaudited PHP's memory_limit
   request termination, because this bug made it possible to reach the
   memory_limit at places that were never meant to be interrupted.
   After a possible exploitation path for Apache 2 servers was
   discovered and a working exploit was created, similar pathes were
   found and added to the proof of concept exploit that allowed
   exploitation of NON Apache 2 servers. (f.e. Apache 1.3.31)

   The idea of the exploit is simple. When PHP allocates a block of
   memory it first checks in the cache of free memory blocks for a block
   of the same size. If such a block is found it is taken from the cache
   otherwise PHP checks if an allocation would violate the memory_limit.
   In that case the request shutdown is triggered through zend_error().
   (PHP < 4.3.7 aborts after the violating memory block is allocated)
   PHP contains several places where such an interruption is unsafe.
   An example for such places are those where Zend HashTables are
   allocated and initialised. This is performed in 2 steps and the
   initialisation step itself allocates memory before important members
   are correctly initialised. An attacker that is able to trigger the
   memory_limit abort within zend_hash_init() and is additionally able
   to control the heap before the HashTable itself is allocated, is
   able to supply his own HashTable destructor pointer.

   Several places within PHP where found where this action is performed
   on HashTables that actually get destructed by the request shutdown.
   One of such places is f.e. within the fileupload code, but is only
   triggerable on Apache 2 servers that are vulnerable to CAN-2004-0493,
   another one is only reachable if variables_order was changed to have
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   the "E" in the end, a third one is within session extension which is
   activated by default but the vulnerability can not be triggered if
   the session functionality is not used. A fourth place is within the
   implementation of the register_globals functionality. Although this
   is deactivated by default since PHP 4.2 it is activated on nearly
   all servers that have to ensure compatibility with older scripts.
   Other places might exist in not default activated or 3rd party
   extensions.

   All mentioned places outside of the extensions are quite easy to
   exploit, because the memory allocation up to those places is
   deterministic and quite static throughout different PHP versions.
   The only unknown entity is the size of the environment vars array.
   But that is usually small and can be bruteforced with some kind
   of binary search algorithm. Additionally this information could
   leak to an attacker through an open phpinfo() page. If the admin
   used php.ini-recommended as configuration basis it is irrelevant
   anyway because the ENV array is not populated in that case.

   Because the exploit itself consist of supplying an arbitrary
   destructor pointer this bug is exploitable on any platform.
   (Except the system runs with non exec heap+stack protection)
   This includes systems running Hardened-PHP <= 0.1.2 because they
   have no protection of the HashTable destructor pointer.

   As a last word it should be said, that an attacker does not need
   to send 8/16/64MB (or whatever the memory_limit is) per attack.
   With POST requests it is quite easy to eat 100 (and more) times
   the amount of sent bytes.

Proof of Concept:

   e-matters is not going to release an exploit for this vulnerability
   to the public.

Disclosure Timeline:

   07. July 2004 - Vendor-sec was informed about the fact that this
                   vulnerability was found
   14. July 2004 - Public Disclosure

CVE Information:

   The Common Vulnerabilities and Exposures project (cve.mitre.org) has
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   assigned the name CAN-2004-0594 to this issue.

Recommendation:

   If you are running PHP with compiled in memory_limit support, it is
   strongly recommended that you upgrade as soon as possible to the
   newest version. Disabling memory_limit within your configuration can
   be considered a workaround, but leaves your site vulnerable to
   memory hungry PHP scripts or POST requests that create huge variables.
   If you are running PHP with Apache <= 2.0.49 ensure that you have the
   fix for CAN-2004-0493 applied.

GPG-Key:

   [http://security.e-matters.de/gpg_key.asc](http://security.e-matters.de/gpg_key.asc)

   pub  1024D/3004C4BC 2004-05-17 e-matters GmbH - Securityteam
   Key fingerprint = 3FFB 7C86 7BE8 6981 D1DA  A71A 6F7D 572D 3004 C4BC

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